STUDENT ID

(M)

No. of pages - 24
No. of printed pages - 23

MID-TERM EXAMINATION (2026-27)
CLASS : X
SUBJECT : SCIENCE (086)

Time Allowed : 3 hours
समय : 3 घंटे

Maximum Marks : 80
अधिकतम अंक : 80

सामान्य निर्देश:

(1) प्रश्न पत्र में 3 खण्डों में 39 प्रश्न हैं। खंड 'क' जीव विज्ञान, खंड 'ख' रसायन विज्ञान एवं खंड 'ग' भौतिक विज्ञान के लिए दिया गया है।

(2) सभी प्रश्न अनिवार्य हैं। कुछ प्रश्नों में आंतरिक विकल्प दिए गए हैं। छात्र से अपेक्षा की जाती है कि वह इनमें से केवल एक प्रश्न का उत्तर दें।

GENERAL INSTRUCTIONS :

1. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

2. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

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SECTION-A

1. The major driving force responsible for upward movement of water in tall trees during day time is: (1)

(a) Root pressure
(b) Transpiration pull
(c) Gravitation pull
(d) ATP as energy source

2. Which of the following is a waste product released by plants during Photosynthesis? (1)

(a) Carbondioxide
(b) Urea
(c) Nitrogen
(d) Oxygen

3. The growth of pollen tubes towards ovules represents: (1)

(a) Hydrotropism
(b) Phototropism
(c) Chemotropism
(d) Geotropism

4. Which gas is mainly responsible for ozone layer depletion? (1)

(a) Oxygen
(b) Nitrogen
(c) Chlorofluorocarbons (CFCs)
(d) Carbon dioxide

5. Which part of the brain receives information from sense organs? (1)

(a) Cerebellum
(b) Cerebrum
(c) Medulla
(d) Spinal cord

6. Which of the following can be considered biodegradable substances? (1)

(a) Plastic bottle
(b) Glass
(c) Vegetable peels
(d) Polythene bag

7. The hormone that prepares the human body to deal with emergency situations is secreted by: (1)

(a) Pancreas
(b) Thyroid gland
(c) Adrenal glands
(d) Pituitary gland

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The following question number 8 and 9 consist of two statements-Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. **Assertion (A):** The opening and closing of stomata is a function of the guard cells. (1)

Reason (R): Swelling and shrinking of guard cells regulates the stomatal opening and closure.

9. **Assertion (A):** Snakes are always primary consumers. (1)

Reason (R): Primary consumers feed directly on plants.

10. Name any two secretions of gastric glands present in the human stomach. Write one important function of each secretion. (2)

11. **Attempt either option (A) or (B)** (2)

(A) Construct a terrestrial food chain comprising four trophic levels. Why do producers occupy only the first trophic level in a food chain?

OR

(B) A school garden has plants, birds, insects, soil, water and air interacting with one another. Identify the biotic and abiotic components of this ecosystem.

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12. A bright flashlight is suddenly focused on a person's eyes. Explain the sequence of events that occurs in response to this stimulus. (2)

13. A student prepares the following flow chart to explain blood circulation in humans: (3)

Vena cava → Right atrium → Right ventricle → Lungs → Left atrium → Left ventricle → Aorta

Using the flow chart, answer the following questions:

(i) Explain the importance of blood passing through the lungs during circulation.

(ii) Which chamber of human heart pumps oxygenated blood to the body? Arteries

(iii) Why is the separation of oxygenated and deoxygenated blood important?

14. State one important function of each of the following plant hormones: (3)

(i) Gibberellin

(ii) Abscisic acid

(iii) Auxin

15. A person consumes a meal rich in carbohydrates, causing an increase in blood glucose level. The body detects this change and releases a specific hormone to bring the glucose level back to normal. This regulation helps maintain a stable internal environment in the body. (4)

(A) Identify the hormone responsible for lowering the increased blood glucose level.

(B) Name the endocrine gland that secretes this hormone.

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Attempt either (C) or (D).

(C) Explain the role of feedback mechanism in maintaining normal blood glucose levels in the body.

OR

(D) If a person's blood glucose level increases after eating but does not return to normal, explain the possible reason for this condition.

16. **Attempt either option (A) or (B)** (5)

(A) (i) Explain the two different pathways by which glucose is broken down to release energy in the human body. Mention the products formed in each pathway.

(ii) Why do aquatic animals have a different breathing rate compared t terrestrial animals? Explain.

(iii) Describe the importance of alveoli in human respiratory system.

OR

(B) (i) What is Dialysis? When it is recommended and how does it work?

(ii) Write one important function of each of the following:

(a) Urinary bladder

(b) Ureter

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SECTION-B

17. Which gas is identified by the 'pop' sound when a burning match stick is brought near it? (1)

(a) Oxygen
(b) Carbon dioxide
(c) Hydrogen
(d) Chlorine

18. The reaction between Sodium sulphate and Barium chloride is called a precipitation reaction because: (1)

(a) Heat is produced.
(b) A gas is evolved.
(c) An insoluble solid is formed.
(d) Water is produced.

19. Which of the following metals is found in the native (free) state? (1)

(a) Sodium
(b) Potassium
(c) Gold
(d) Calcium

20. When a metal hydrogen carbonate reacts with an acid, the products formed are: (1)

(a) Salt and water
(b) Salt, water and hydrogen gas
(c) Salt, water and carbon dioxide
(d) Metal oxide and water

21. Which of the following can be used to treat a bee sting? (1)

(a) Vinegar
(b) Baking soda solution
(c) Hydrochloric acid
(d) Lemon juice

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Question No. 22 consists of two statements- Assertion (A) and Reason (R). Answer this question by selecting the appropriate option given below:

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

22. **Assertion (A):** Burning of magnesium ribbon in air is a combination reaction. (1)

Reason (R): Magnesium combines with oxygen to form magnesium oxide.

23. How is the concentration of hydronium ions (H_3O^+) affected when a solution of an acid is diluted? What happens to its pH? (2)

24. Silver chloride is always preferred to store in dark coloured bottles. Explain the reason. (2)

25. **Attempt either option (A) or (B)** (3)

(A) Copper sulphate crystals look blue but turn white on heating in a test tube.

(i) Why do the crystals change colour on heating?

(ii) What happens if 2-3 drops of water are added to the white powder obtained?

(iii) Write the chemical formula for hydrated copper sulphate.

OR

(B) A substance "Z", which is a member of the salt family, is used for disinfecting drinking water.

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(i) Identify substance "Z" and write its chemical name.

(ii) How is this substance prepared from a product of the chlor-alkali process?

(iii) Write the balanced chemical equation for its preparation.

26. What is thermite reaction? State one of its uses and write the balanced chemical equation for it. (3)

27. A student performed an experiment in his school laboratory. He took about 20 ml of blue-coloured copper sulphate ($CuSO_4$) solution in a test tube and dipped a clean, greyish iron nail into it. He left the setup undisturbed for about 3 hours. When he returned, he noticed that the blue colour of the solution had faded significantly and turned light green. Furthermore, a reddish-brown coating had formed on the surface of the iron nail. (4)

(A) Identify the type of chemical reaction that took place in the test tube.

(B) Write a balanced chemical equation for the above reaction.

Attempt either (C) or (D).

(C) Why did the blue colour of the solution fade and change to light green?

OR

(D) What would happen if the student dipped a copper wire into iron sulphate ($FeSO_4$) solution? Explain your with reason? (5)

28. **Attempt either option (A) or (B).**

(A) (i) Show the formation of magnesium chloride by the transfer of electrons. State the ions present in this compound. (Atomic no.: Mg=12, Cl=17)

(ii) Why do ionic compounds not conduct electricity in their solid state, but become excellent conductor when dissolved in water?

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OR

(B) (i) What are amphoteric oxides? Write balanced chemical equations of reactions of an amphoteric oxide with both an acid and a base.

(ii) Generally, hydrogen gas is not evolved when a metal reacts with dilute HNO_3, why?

SECTION-C

29. For a concave mirror, the magnification will be (–1) when the object is placed at: (1)

(a) at focus F (b) beyond C

(c) Between C and F (d) At centre of curvature

30. The image formed on the retina of human eye is: (1)

(a) Virtual and erect (b) Real and inverted

(c) Virtual and inverted (d) Real and erect

31. The power of concave lens is always: (1)

(a) Positive (b) Negative

(c) Zero (d) Infinite

32. The near point and far point of human eye are: (1)

(a) 15 cm and 25 cm respectively (b) 25 cm and infinite respectively

(c) 25 cm and 15 cm respectively (d) infinite and 25 cm respectively

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Question No. 33 consists of two statements- Assertion (A) and Reason (R). Answer this question by selecting appropriate option given below:

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

33. **Assertion (A):** Power of accommodation remains constant throughout life. (1)

Reason (R): The flexibility of the eye lens decreases with age.

34. An astronaut in space finds sky to be dark. Explain reason for his observation. (2)

OR

Explain 'dispersion of white light'? Name the colour that bends the most and least when white light is passed through a glass prism.

35. Calculate the image distance and height of image formed when an object of height 10 cm is placed 25 cm away from the optical centre of a converging lens of focal length 15 cm. (3)

36. Explain the formation of rainbow in the sky describing three optical phenomena involved in their correct sequence. (3)

37. (i) Explain the absolute refractive index of a medium. Write its formula. (3)

(ii) Why does a light ray bend when it travels obliquely from air into glass?

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38. A 15-year-old student studying in Class 10 recently noticed that while sitting on the last bench of his classroom, he struggled to read the formulas written on the blackboard clearly, even though he could read his textbook without any strain. His class teacher advised him to visit an eye specialist. The doctor examined his eyes and prescribed him spectacles with corrective lenses. (4)

(A) Identify the defect the student is suffering from and state where the image of a distant object is formed in such an eye.

(B) Name the type of lens prescribed by the doctor to correct his vision.

Attempt either (C) or (D).

(C) List two causes responsible for this defect in his vision.

OR

(D) Is there any change in the near point and far point of an eye affected by such a vision defect? Explain.

39. **Attempt either option (A) or (B)** (5)

(A) (i) Identify the type of mirror used in the following situations. Also, state on reason why they are preferred for this purpose.

(a) Rear view mirror in vehicles.

(b) head light of cars.

(c) mirror used by dentists.

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(ii) Explain the following in context of a spherical mirror.

(a) Focal length

(b) Principal focus

OR

(B) (i) An object 4 cm in height is placed at a distance of 25 cm in front of a concave mirror of focal length 15 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image? Also find the size of the image.

(ii) "The magnification (m) produced by a spherical mirror is found to be (–0.5)". What inferences can be drawn from the above statement about nature and size of image?

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